

1)Implementation of FIND -S ALGORITHM:

CODE:

```
# Dataset: [Age, Experience, Salary, Hire]
```

```
data = [  
    ['Young', 'Low', 'Low', 'No'],  
    ['Young', 'High', 'High', 'Yes'],  
    ['Middle', 'High', 'Medium', 'Yes'],  
    ['Old', 'High', 'High', 'Yes']  
]
```

```
# Initialize Specific Hypothesis
```

```
S = ['0'] * (len(data[0]) - 1)
```

```
# Find-S Algorithm
```

```
for row in data:
```

```
    x, y = row[:-1], row[-1]
```

```
    if y == 'Yes':
```

```
        if S[0] == '0':
```

```
            S = x.copy()
```

```
        else:
```

```
            for i in range(len(S)):
```

```
                if S[i] != x[i]:
```

```
                    S[i] = '?'
```

```
print("Final Specific Hypothesis (S):", S)
```

OUTPUT:

```
# Dataset: [Age, Experience, Salary, Hire]
data = [
    ['Young', 'Low', 'Low', 'No'],
    ['Young', 'High', 'High', 'Yes'],
    ['Middle', 'High', 'Medium', 'Yes'],
    ['Old', 'High', 'High', 'Yes']
]

# Initialize Specific Hypothesis
S = ['0'] * (len(data[0]) - 1)

# Find-S Algorithm
for row in data:
    x, y = row[:-1], row[-1]

    if y == 'Yes':
        if S[0] == '0':
            S = x.copy()
        else:
            for i in range(len(S)):
                if S[i] != x[i]:
                    S[i] = '?'

print("Final Specific Hypothesis (S):", S)

... Final Specific Hypothesis (S): ['?', 'High', '?']
```